

ANSYS HFSS Student Edition Model Construction Guide

Overview

This document provides the complete procedure for constructing, simulating, and post-processing the proposed HFSS model using **Ansys Electronics Desktop (HFSS Student Edition)**. The model is designed for a **60 mm × 60 mm** tissue footprint and is optimized to remain within the mesh limitations of the Student Edition while allowing a parametric sweep of subcutaneous fat thickness.

Phase 1 – Environment Setup & Variable Mapping

1.1: Create a New Project

- 1. Open **Ansys Electronics Desktop**.
- 2. Create a **New HFSS Design**.

1.2: Define Design Variables

Navigate to:

HFSS → Design Properties → Add

Create the following local variables.

Variable	Value	Description
fat_thickness	5 mm	Baseline variable for parametric sweep
muscle_thickness	15 mm	Muscle layer thickness
skin_thickness	1.5 mm	Skin layer thickness
tissue_width	60 mm	Overall tissue width and length
superstrate_thickness	1 mm	Superstrate thickness
patch_length	14 mm	Patch antenna length
patch_width	14 mm	Patch antenna width

1.3: Set Solution Type

Right-click the design name in the Project Tree.

Select:

Solution Type → Terminal

Phase 2 – Biological Tissue Model Construction

Construct three stacked biological tissue layers.

2.1 Muscle Layer

Create a rectangular box.

Geometry

Parameter	Value
Material	Muscle (or Custom)
Relative Permittivity (ϵ_r)	54.5
Conductivity (σ)	0.94 S/m
Position	(-tissue_width/2, -tissue_width/2, 0 mm)
X Size	tissue_width
Y Size	tissue_width
Z Size	muscle_thickness

Evaluated position:

- X = -30 mm
- Y = -30 mm
- Z = 0 mm

2.2 Subcutaneous Fat Layer

Create a second box directly above the muscle.

Geometry

Parameter	Value
Material	Fat (or Custom)
Relative Permittivity (ϵ_r)	5.3
Conductivity (σ)	0.04 S/m
Position	(-tissue_width/2, -tissue_width/2, muscle_thickness)
X Size	tissue_width
Y Size	tissue_width
Z Size	fat_thickness

Evaluated position:

- X = -30 mm
- Y = -30 mm
- Z = 15 mm

This layer will be varied parametrically.

2.3 Skin Layer

Create the third biological layer.

Geometry

Parameter	Value
Material	Skin (or Custom)
Relative Permittivity (ϵ_r)	41.4
Conductivity (σ)	0.69 S/m
Position	(-tissue_width/2, -tissue_width/2, muscle_thickness + fat_thickness)
X Size	tissue_width
Y Size	tissue_width
Z Size	skin_thickness

Baseline evaluated position:

- X = -30 mm
- Y = -30 mm
- Z = 20 mm

Phase 3 – Structural Component Modeling

3.1 Ground Plane

Create a rectangle.

Geometry

Parameter	Value
Position	(-tissue_width/2, -tissue_width/2, 0 mm)
Axis	Z
X Size	tissue_width
Y Size	tissue_width

Assign Boundary:

Perfect E

3.2 Internal Sensor Cylinder

Create a cylinder.

Geometry

Parameter	Value
Center Position	(0, 0, 17.5 mm)
Axis	Z
Radius	0.5 mm
Height	2 mm
Number of Segments	0

This places the sensor approximately at the center of the fat layer.

3.3 External Patch Antenna

Create a rectangle.

Geometry

Parameter	Value
Position	$(-\text{patch_length}/2, -\text{patch_width}/2, \text{muscle_thickness} + \text{fat_thickness} + \text{skin_thickness})$
Axis	Z
X Size	patch_length
Y Size	patch_width

Baseline evaluated position:

- X = -7 mm
- Y = -7 mm
- Z = 21.5 mm

Assign Boundary:

Perfect E

3.4 Ceramic Superstrate

Create a dielectric box.

Geometry

Parameter	Value
Material	Alumina (or Custom Ceramic)
Position	$(-30 \text{ mm}, -30 \text{ mm}, \text{muscle_thickness} + \text{fat_thickness} + \text{skin_thickness})$
X Size	60 mm
Y Size	60 mm
Z Size	2 mm

Baseline evaluated position:

- X = -30 mm
- Y = -30 mm
- Z = 21.5 mm

3.5 Lumped Port Feed Bridge

Create a rectangle.

Geometry

Parameter	Value
Position	(0, 0, muscle_thickness + fat_thickness/2)
Axis	Y
Width (X Size)	0.5 mm
Height (Z Size)	-(muscle_thickness + fat_thickness/2)

Baseline evaluated values:

- Feed Position = (0, 0, 17.5 mm)
- Feed Extension = -17.5 mm

Assign Excitation:

Lumped Port

Reference Impedance:

50 Ω

3.6 Radiation Region

Create the simulation region.

Region Settings

Padding Type:

Absolute Offset

Padding in all six directions:

248.13895781638 mm

Assign Boundary:

Radiation

This corresponds approximately to a quarter-wavelength clearance at 600 MHz.

Phase 4 – Analysis and Parametric Sweep

4.1 Solution Setup

Navigate to:

Analysis → Add Solution Setup

Use the following settings.

Parameter	Value
Center Frequency	600 MHz
Maximum Passes	10
Maximum Delta S	0.02

These settings are optimized for the Student Edition mesh limitations.

4.2 Frequency Sweep

Right-click the created setup.

Select:

Add Frequency Sweep

Use the following configuration.

Parameter	Value
Sweep Type	Fast (or Interpolating)
Start Frequency	200 MHz
Stop Frequency	800 MHz
Step Size	1 MHz

This produces 601 frequency samples.

4.3 Parametric Sweep

Navigate to:

Optimetrics → Add → Parametric

Choose the variable:

fat_thickness

Configure the sweep.

Parameter	Value
Sweep Type	Linear Step
Start	5 mm
Stop	15 mm
Step	2.5 mm

The simulation automatically evaluates five tissue configurations:

- 5 mm
- 7.5 mm
- 10 mm
- 12.5 mm
- 15 mm

Under the **Options** tab, enable:

Save Fields and Mesh

Phase 5 – Validation and Simulation

Validation

From the toolbar select:

HFSS → Validation Check

Verify that:

- No geometry errors exist.
- Material assignments are complete.
- Boundary conditions are valid.
- Port definitions are correct.
- Radiation region is properly assigned.

Run Simulation

Navigate to:

Optimetrics → Parametric Setup

Right-click and select:

Analyze

HFSS will execute all parameter combinations automatically.

Phase 6 – Post-Processing and Result Generation

After the simulation completes, generate the following engineering plots.

6.1 S-Parameter Plot

Navigate to:

Results → Create Terminal Solution Data Report → Rectangular Plot

Configuration:

Setting	Value
Category	Terminal S Parameter
Quantity	S(1,1)
Function	dB

Create the report.

6.2 VSWR Plot

Navigate to:

Results → Create Terminal Solution Data Report → Rectangular Plot

Configuration:

Setting	Value
Category	Terminal VSWR
Quantity	VSWR(1)
Function	None

Create the report.

6.3 Input Impedance Plot

Navigate to:

Results → Create Terminal Solution Data Report → Rectangular Plot

Real Component

Setting	Value
Category	Terminal Z Parameter
Quantity	Z(1,1)
Function	Re

Click **Add Trace**.

Imaginary Component

Use the same settings except:

Setting	Value
Function	Im

Create the report containing both traces.

6.4 2D Far-Field Gain Plot

Navigate to:

Results → Create Far Fields Report → Rectangular Plot

Configuration:

Setting	Value
Category	Gain
Quantity	GainTotal
Function	dB

Under the **Families** tab, select the desired angular cut:

Create the report.

6.5 3D Radiation Pattern

Navigate to:

Results → Create Far Fields Report → 3D Polar Plot

Configuration:

Setting	Value
Category	Gain
Quantity	GainTotal
Function	dB

Create the report to generate the complete three-dimensional radiation pattern.

Expected Outputs

The simulation should generate the following result set:

- S11 (Return Loss) vs Frequency
- VSWR vs Frequency
- Input Resistance (Real Impedance)
- Input Reactance (Imaginary Impedance)
- 2D Far-Field Gain Pattern
- 3D Radiation Pattern

Additionally, the parametric sweep will evaluate all five fat-layer thickness configurations, enabling comparison of antenna performance as the subcutaneous fat thickness varies.